



Ques: Segregate 0s and 1s

M-1

arr[] = [0, 0, 0, 0, 0, 1, 1, 1, 1, 1]

zeros = 0 1 2 3 4 5

ones = 6 7 8 9 10

(Lecture 21)

By - Raghav Sir

Summary Notes

Ques: Segregate 0s and 1s

Method-2



Earn₹s

[0, 1, 0, 1, 0, 0, 1, 1, 1, 0]

0 0 0 0 0 1 1 1 1 1

i
j

Ques: Segregate 0s and 1s

```
int i = 0, j = arr.size() - 1;
while(i < j){
    if(arr[i] == 0) i++;
    if(arr[j] == 1) j--;
    if(arr[i] == 1 && arr[j] == 0){
        swap(arr[i], arr[j]);
        i++;
        j--;
    }
}
```

1 0 1 0 1 0 0

0 0 0 1 0 1 1
 j i

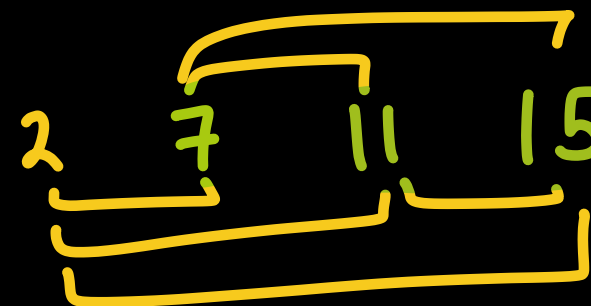
↗ The Problem ↖



Ques: Two Sum

nums = [2, 7, 11, 15], target = 22

ans = {1, 3}



tar = 22

nums = {4, 3, 2}, tar = 6

ans = {0, 2}

Ques: Missing in Array

$arr = \{6, 4, 1, 5, 3, 0\}$

Missing no = 2

M-1 : Nested Loops

M-2 : Sorting

$\{0, 1, 2, 3, 4, 5, 6\}$

$n = 6$

```
for (i = 0 to n) {  
    bool flag = false  
    for (int ele : arr)  
    {  
        if (i == ele)  
        {  
            flag = true  
            break  
        }  
    }  
}
```

```
} if flag == false → return  
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```

Method 3 $\Rightarrow$ Extra Space

Ques: Missing in Array

arr = { 5, 4, 1, 6, 0, 2 }

missing no = 3

n = 6

TC = $O(n)$

flag =

0	1	2	3	4	5	6
T	T	T	F	T	T	T

AS = $O(n)$

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Method-4 : Maths

Ques: Missing in Array



Earn₹s

$$\text{arr} = \{ 5, 4, 1, 6, 0, 2 \} \quad \text{missing no} = 3$$

$$n = 6$$

$$\text{arraySum} = 18$$

$$\text{zeroToNSum} = \frac{n(n+1)}{2} = \frac{6 \cdot 7}{2} = 21$$

$$21 - 18 = 3$$

Code → HW

$$TC = O(n)$$

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Ques: Wave Array

Ques: 66. Plus One

$$\text{arr} = \{9, 9, 9\}$$

$$\text{ans} = \{1, 0, 0, 0\}$$

$$\begin{array}{r} \overleftarrow{1} \quad 1 \\ 9 \quad 9 \quad 9 \\ + \quad 1 \\ \hline 0 \quad 0 \quad 0 \end{array}$$

$$\text{carry} = 1 \quad 1 \quad 1 \quad 1$$

$$\text{ans} = \{0, 0, 0, 1\}$$

↓
rev

Ques: 66. Plus One

$$\text{arr} = \{1, 2, 9, 9\}$$

$$\text{ans} = \{1, 3, 0, 0\}$$

$$\text{arr} = \{9, 1, 0, 9\}$$

$$\text{ans} = \{9, 1, 1, 0\}$$

$$\begin{array}{r} \leftarrow \\ 0 \ 1 \ 1 \\ 1 \ 2 \ 9 \ 9 \\ + \ 1 \\ \hline 1 \ 3 \ 0 \ 0 \end{array}$$

$$\text{carry} = \cancel{1} \cancel{1} \\ \phantom{\text{carry}} 1 \ 0 \ 0$$

$$\text{ans} = \{0, 0, 3, 1\}$$

↓
reverse